COALITION BARGAINING IN NATURAL LANGUAGE

LLM-negotiated joint programmes of government

Simulating how parties turn rival manifestos into an agreed governing mandate

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THE PUZZLE

Forming a coalition means writing a governing mandate

179

PDF pages

The CDU, CSU and SPD coalition agreement of 7 February 2018, negotiated after the September 2017 election

Most elections give no party a majority. Parties get into power only by agreeing a joint programme with others: a mandate nobody put to the voters.

- Constrains ministerial autonomy
- Structures legislative scrutiny
- Links campaign promises to what governments actually do

WHERE THE LITERATURE STANDS

Three traditions, one missing piece

Formal theory

From Riker and Gamson to Baron–Ferejohn and Laver–Shepsle: dividing a prize, or picking a point in a low-dimensional policy space.

Formation studies

Empirical work on the time from election to new government shows that bargaining delays are endemic.

Agreement studies

Coalition agreements analysed after they are signed, as tools for constraining ministers and tracking pledges.

THE GAP No one models the negotiations that turn rival manifestos into an agreed text

THE MOVE

Replace coordinates with texts

Element	Analytical models	This paper
Party preferences	Ideal points in a low-dimensional space	Full manifesto texts, in the original language
Outcome	A policy point	The full text of an agreed programme
Status quo	A reversion point	A reversion programme, R: policy in force under the caretaker
Bargaining	Take-it-or-leave-it offer	Drafting, amendment, veto, investiture
Information	Parameterized ***risk*** (sometimes)	Interpretive ***uncertainty***

THE KEY ASSUMPTION

Negotiators face interpretive uncertainty, not quantifiable risk

A *formateur's* first offer rests on its reading of partners' priorities, buried in long texts. That reading can be wrong, and *a* ***wrong reading leads to bad forecasts and failed attempts***.

Reasons for failure are logged, and the failure history passed on to the next *formateur*.

THE AGENTS

Each party is an LLM agent holding three things

Its seat share
From the actual election result

Its full manifesto
Unreduced, in the original language

The reversion programme, R
Policy actually in force under the caretaker government, not the outgoing coalition's unmet promises

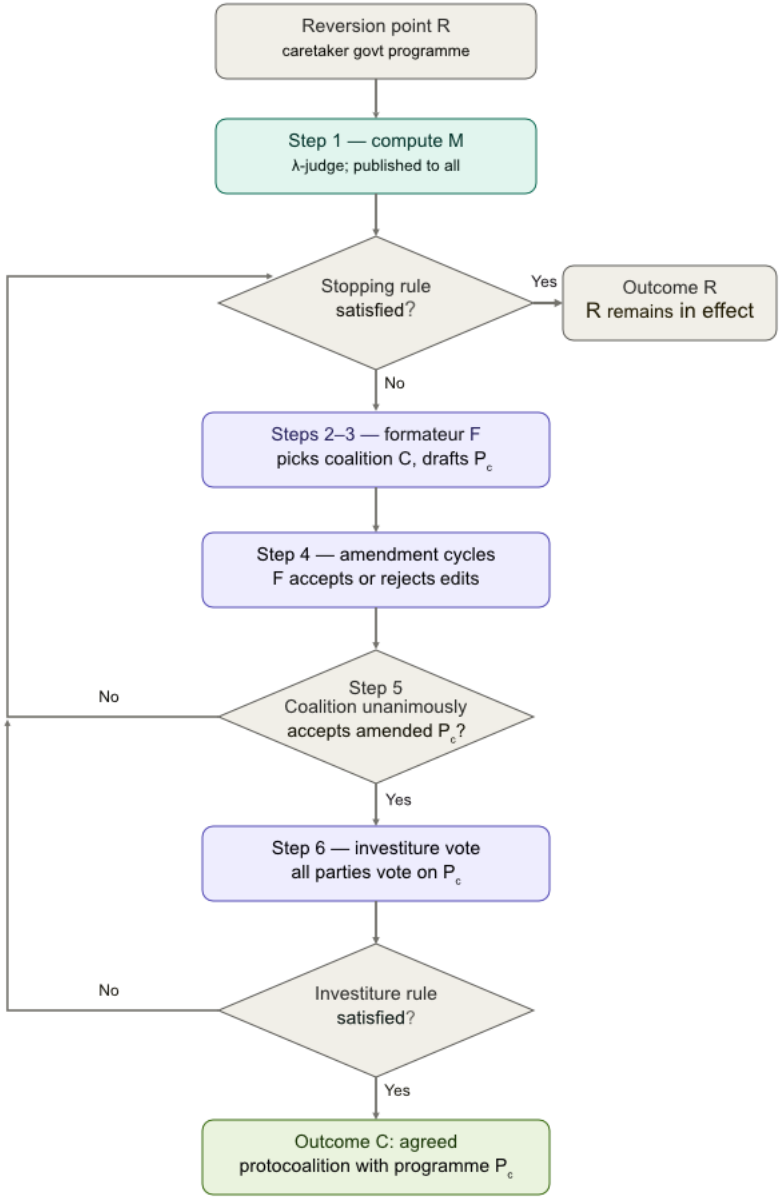
Baseline assumptions

- Sincere: no agent misrepresents its preferences
- Policy only: no value placed on cabinet posts, therefore …
- … not a closed model of formation: portfolios are not allocated
- Hence "protocoalitions", not coalitions

THE PROCEDURE

Six steps, looping until a government is invested

1. Score how similar every pair of manifestos is, and publish the scores to all parties
2. A protocol names a formateur
3. Formateur picks partners and drafts a programme
4. Partners propose amendments over a fixed number of rounds
5. Every protocoalition member must approve; any one can veto
6. Investiture vote in the full legislature



INSIDE THE NEGOTIATION

Draft, amend, ratify, invest

Draft

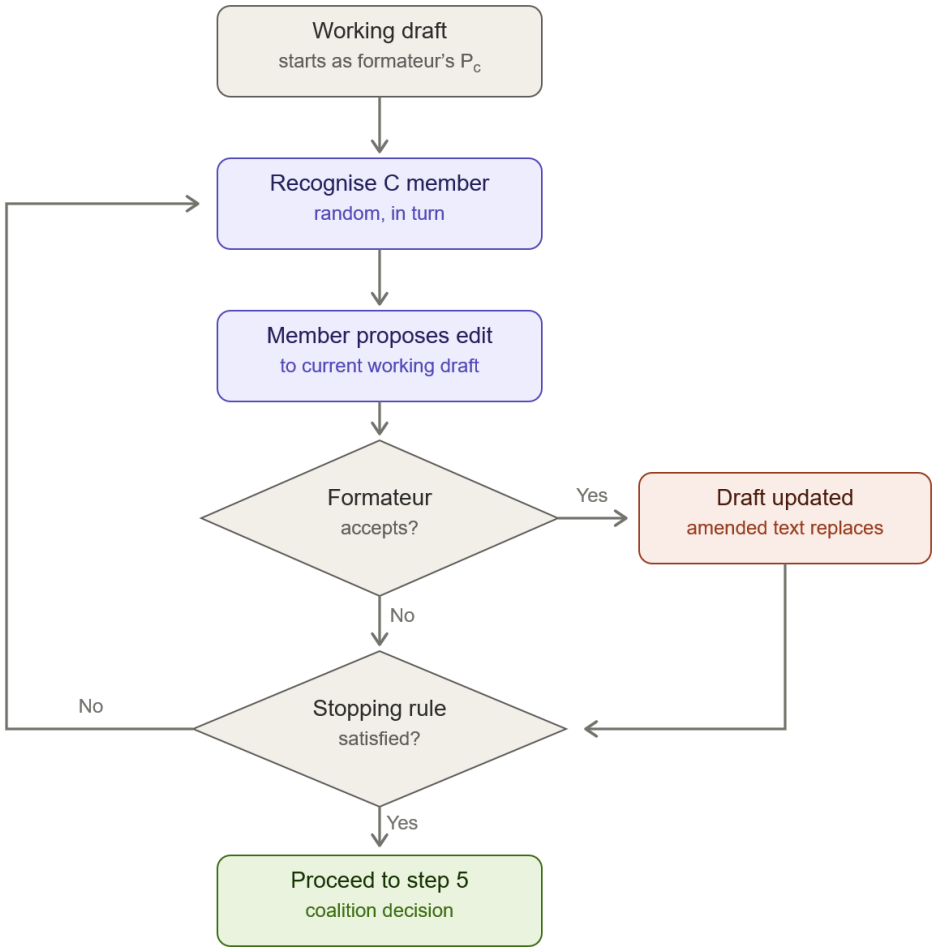
Formateur writes a programme as close to its own manifesto as possible, provided it prefers it to R, judges that partners do too, and expects it to pass investiture.

Amend

Partners, in random turn, approve, propose a replacement section, or flag an obstacle. Formateur accepts or rejects each edit whole.

Ratify and invest

Unanimous protocoalition approval, then a floor vote. Outsiders are reminded that backing this programme may cost them a later chance to help write one.



Step 4: the amendment loop

MEASUREMENT

Texts are not points, so an LLM judges similarity

A separate LLM "λ-judge" rates how similar two texts are, from 0 to 1. It is used three ways:

Before bargaining	Matrix M of manifesto similarities, to guide partner choice
Policy payoffs	How far the agreed programme accommodates each party's manifesto
Accuracy	Simulated programme versus the real one

Does M track expert judgement? Rank correlation with Chapel Hill Expert Survey left–right similarities:

0.94	0.85	0.82
Germany	Slovenia	Denmark

CASES AND DESIGN

Three elections of rising complexity

Germany 2005

31

possible coalitions of 5 party blocs

Majority investiture. Real outcome: CDU/CSU–SPD grand coalition. Test: a cordon sanitaire on Left-PDS.

Inside model training data: calibration/debuggung case

Slovenia 2026

127

possible coalitions of 7 parties

Majority investiture. Real outcome: SDS–NSi-SLS-Fokus–Demokrati minority. Test: no GS–SDS coalition.

After model training cutoff

Denmark 2026

4,095

possible coalitions of 12 parties

Negative parliamentarism. Real outcome: S–RV–SF–M. Test: no SF–V cabinet.

After model training cutoff

3 protocols × 2 constraint conditions × 50 repetitions = **300 runs per country, 900 in all**

THE STARTING MAP

Who is close to whom, and to the status quo

Darker cells mean more similar manifestos. "Peer mean" measures a party's centrality in the system.

Germany

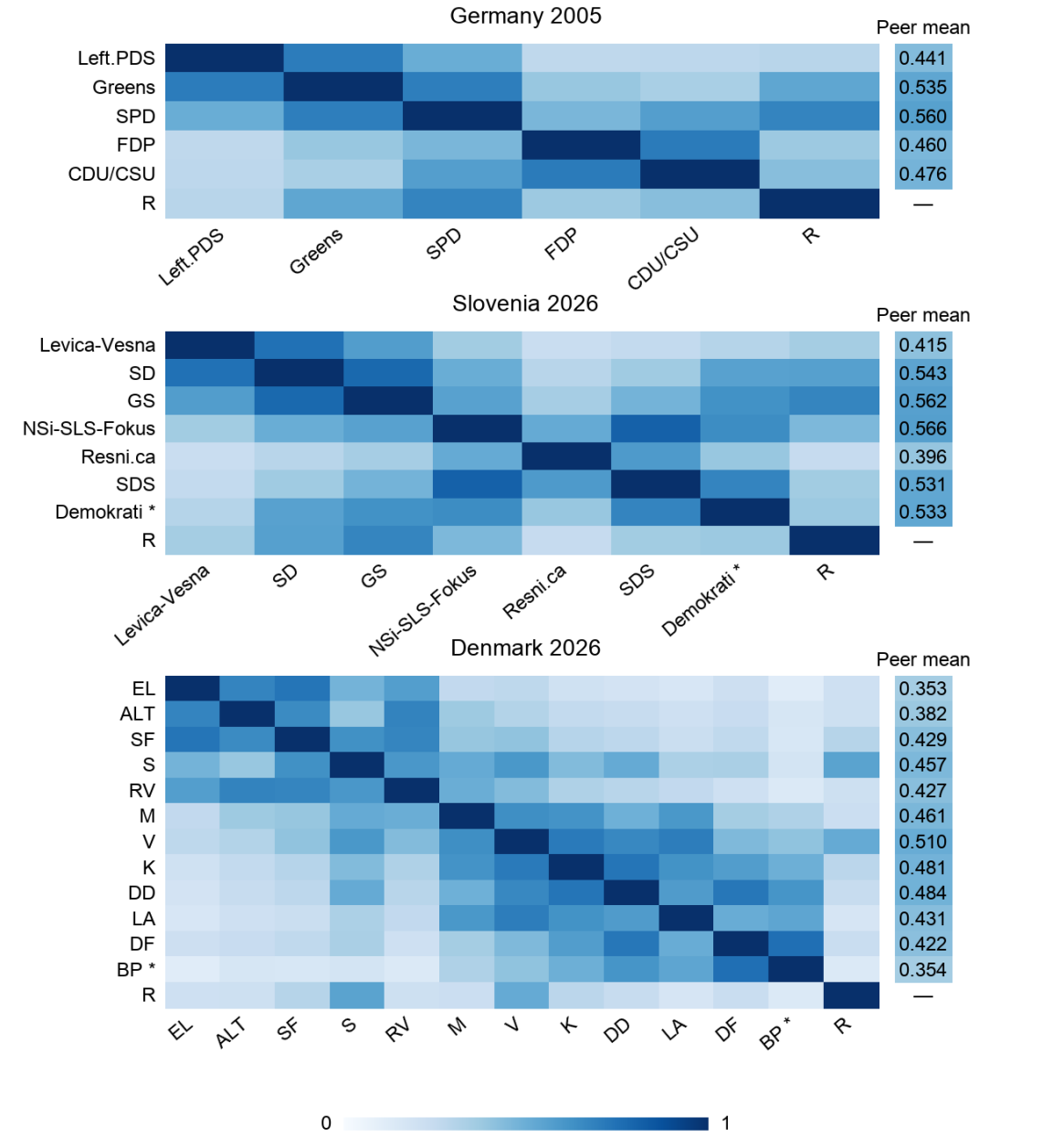
SPD is both the most central party and the closest to R

Slovenia

GS, which led the caretaker cabinet, is closest to R

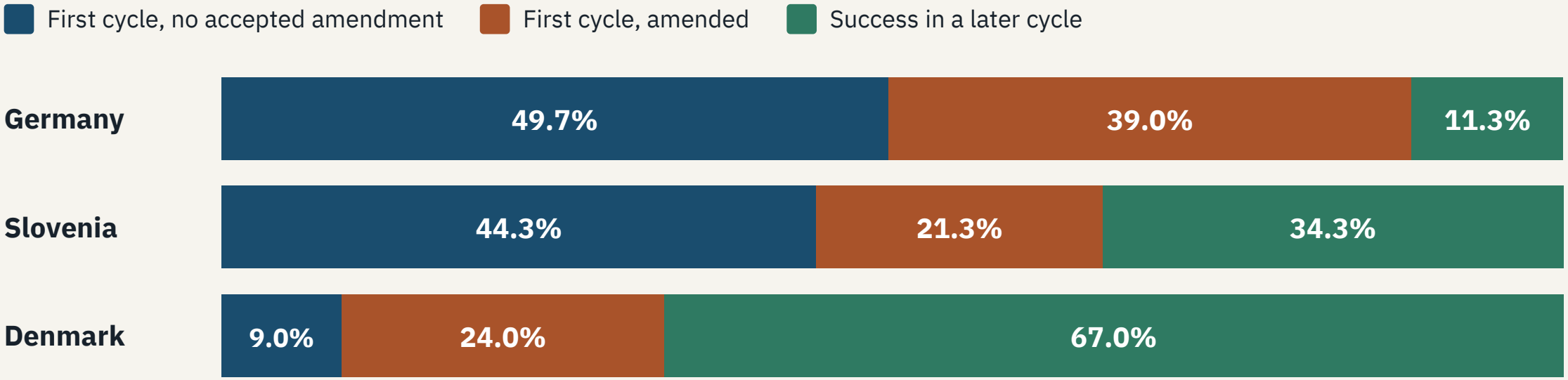
Denmark

Venstre is the most central; S and V are closest to R



HEADLINE RESULT

All 900 runs invested a programme, typically not on the first draft



Share of successful runs, 300 per country. No run fell back to the caretaker status quo. Amendments surface objections that would otherwise sink the deal.

WHAT A NEGOTIATION LOOKS LIKE

Two German runs: a VAT deal and a health red line

CDU/CSU DRAFTS

A VAT increase to fund lower non-wage labour costs

SPD OBJECTS

Asks for its removal. CDU/CSU rejects: financing not specified

SPD COUNTERS

Two percentage point increase from January 2006, targeted compensation, no further rise

DEAL

Accepted in round three; survives verbatim into the invested programme

A different run: CDU rejecting an SPD amendment on health financing .

> *"This fourth version, however, directly violates the red line previously stated on healthcare financing … the two remaining rounds still permit a targeted correction."*

A stated justification, not a window into the model's decision process.

WHERE ATTEMPTS FAIL

Deals mostly die on the floor, not at the table

523
failed the investiture vote

31
vetoed by a protocoalition member

26
other German failures: excluded party pivotal, or formateur declined

Out of 1,480 formation attempts. The usual error: formateurs misforecast outside support. Expected outside "aye" votes that actually came:

Germany 10 of 41 **Slovenia** 162 of 375 **Denmark** 637 of 1,322

PREDICTED PROTOCOALITIONS

Which protocoalitions pass investiture?

Germany

10 cabinets formed, of 31 possible

Cabinet	% runs	λ
CDU/CSU + SPD †	82.7	0.83
SPD + Greens + Left-PDS	6.0	0.54
SPD + Greens	4.3	0.66
CDU/CSU + FDP + Greens	2.0	–

Slovenia

26 cabinets formed, of 127; 71% minority

Cabinet	% runs	λ
GS + NSi-SLS-Fokus + SD	22.0	0.67
SDS + NSi-SLS-Fokus + Demokrati †	19.0	0.89
GS + SD + Demokrati	8.3	0.63
GS + SDS	7.7	0.84

Denmark

50 cabinets formed, of 4,095; 89% minority

Cabinet	% runs	λ
S + RV + SF + V	11.0	0.82
S + V	10.7	0.78
K + LA + M + V	9.3	0.70
K + LA + V	9.0	0.67
S + RV + SF + M †	3.0	0.78

† the cabinet that actually formed. λ: similarity of the simulated programme to the real one. Denmark's real-cabinet λ rests on only 9 runs.

PARTNERS VERSUS PROGRAMMES

Right partners → realistic text.
Finding the right partners is the hard part.

Case	Real cabinet formed	λ, real cabinet	λ, other cabinets
Germany	83% of runs	0.83	0.64
Slovenia	19% of runs	0.89	0.74
Denmark	3% of runs	0.78 (9 runs)	0.82 (top cabinet)

0.98 Faithful summaries of the real Merkel I programme, scored against it

0.73 Outgoing SPD–Green programme against the incoming CDU–SPD one

All three post-cutoff cases (incl Sweden) result in **minority coalitions**, which no current analytical model can handle

PROPOSAL POWER

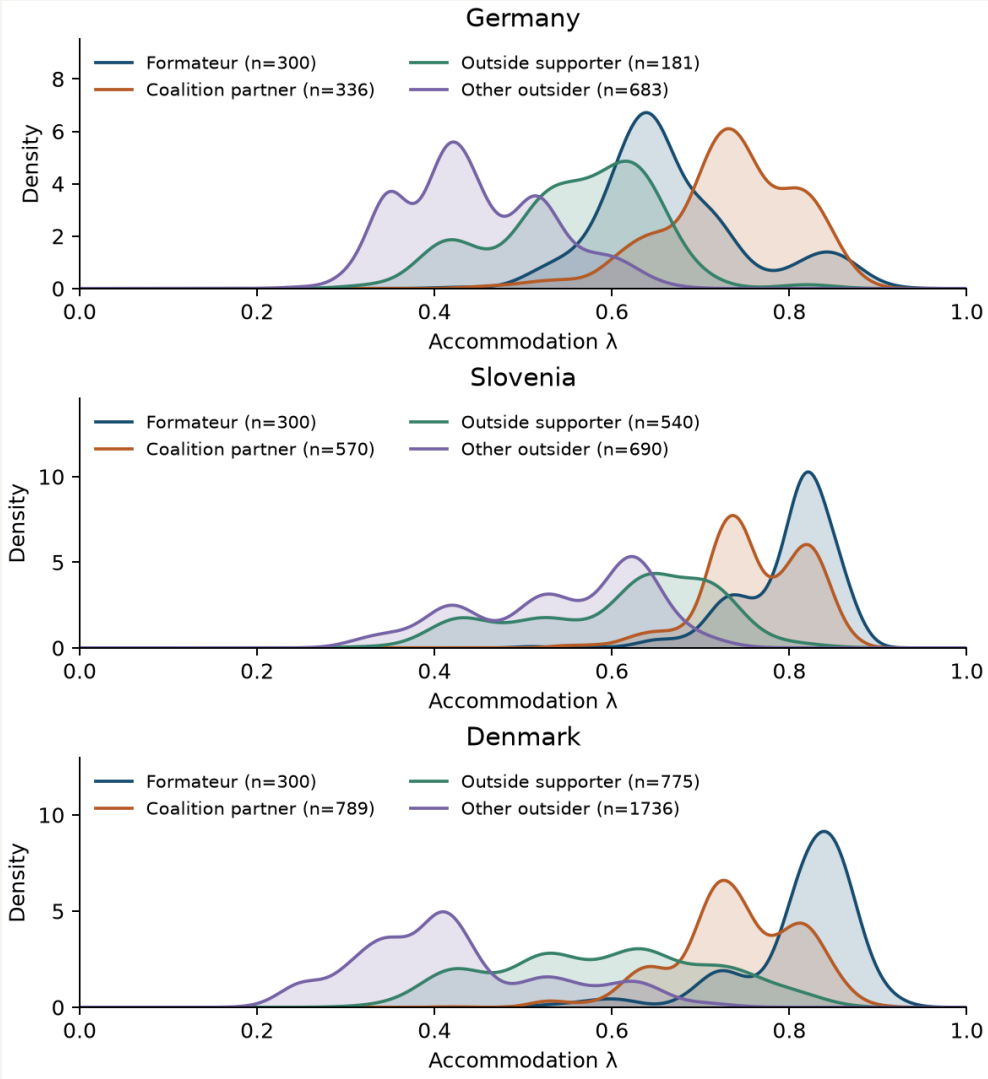
Is there a formateur advantage?

In Slovenia and Denmark, policy payoffs follow the order Baron–Ferejohn predicts: formateur, then partners, then outside supporters, then other outsiders.

Germany is the exception

Formateur	CDU/CSU λ	SPD λ
CDU/CSU (234 runs)	0.64	0.75
SPD (14 runs)	0.62	0.78

SPD, the most central party and the closest to R, comes out ahead either way.



EXPERIMENTAL ARMS

Red lines and recognition rules both shape the outcome

Adding a declared exclusion raises recovery of the real cabinet

+17	**Germany**: cordon sanitaire on Left-PDS
+9	**Slovenia**: no GS–SDS coalition
+3	**Denmark**: no SF–V cabinet

Percentage-point change in runs forming the real cabinet

Who proposes first matters, where first proposals succeed

−38

Germany: Baron–Ferejohn recognition forms the grand coalition about 38 points less often than rules putting CDU/CSU first.

No clear effect in Slovenia or Denmark.

A BENCHMARK

Negotiated programmes versus a top-down "Oracle"

BOTTOM-UP · 600 PROGRAMMES

The full behavioural model: formateur drafts, partners amend, the formateur accepts or rejects, partners approve

TOP-DOWN · 600 PROGRAMMES

One LLM, given the blinded manifestos and seat shares, told to integrate the partners' priorities into about 1,000 words

Both land about equally close to the real programme. Negotiation is as realistic as direct drafting, direct drafting, but it also shows ***how the agreed text was reached.***

LIMITS AND NEXT STEPS

Why is choosing partners harder than drafting?

1

Office perks are missing

Utility = α·perks + (1 − α)·policy, and the model sets α to zero. Outsiders may back cabinets that, wanting portfolios, they would really vote down.

2

Information is thin at selection

Formateurs choose partners from the summary matrix M. Giving Danish formateurs every manifesto (154,812 words) showed no clear gain.

3

The tasks differ

LLMs may simply be better at drafting compromise text than at searching 2,048 possible protocoalitions for a winning route.

TAKEAWAYS

Coalition bargaining, modelled as what it is: drafting text

Texts in, text out

Full manifestos and a real status quo replace points in a policy space.

Two distinct hurdles

Agreeing a text with partners is not the same as winning the floor, and most failures happen on the floor.

Realistic mandates

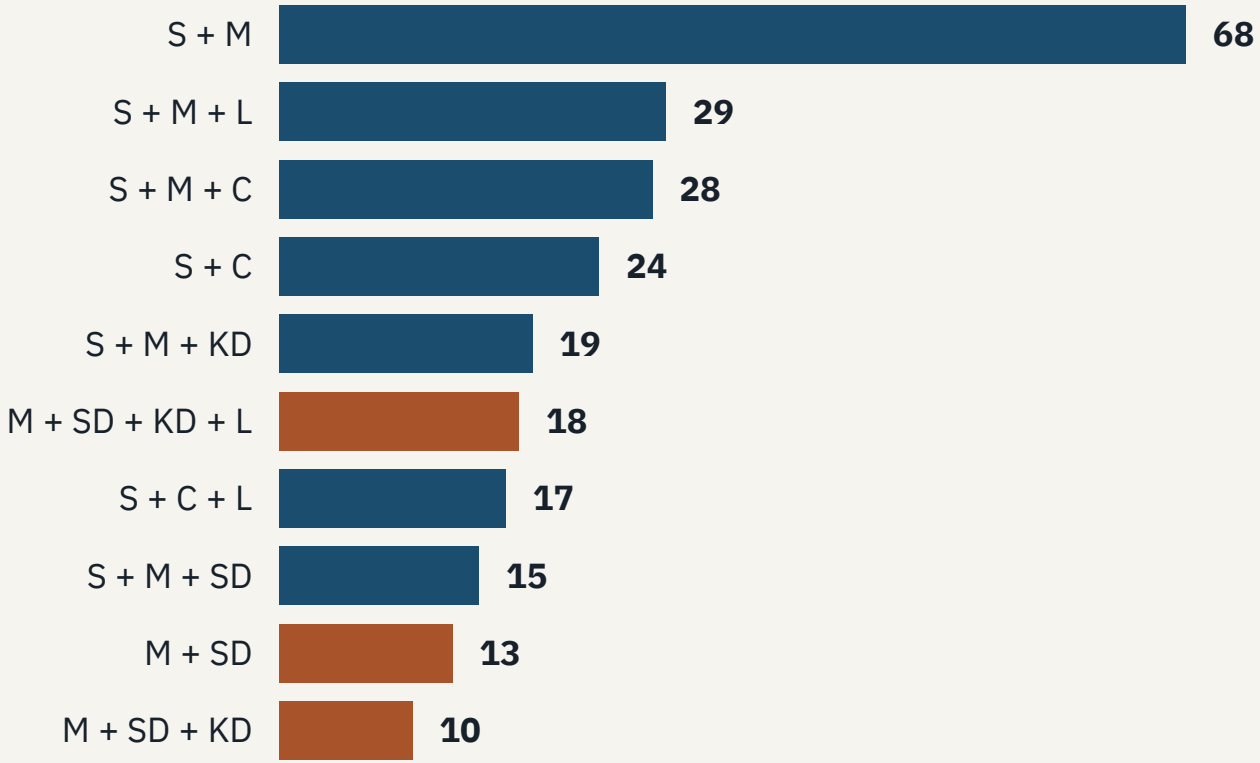
With the right partners, simulated programmes closely resemble real ones in Germany and Slovenia.

Next: perks and partners

Add office payoffs and better information for choosing partners, and test on Sweden.

ADDENDUM · AN OUT-OF-SAMPLE TEST

Sweden 2026: predicted before the outcome



The ten most frequent simulated cabinets, out of 300 runs. Blue bars include the Social Democrats; orange bars do not.

S appears in 243 runs and leads 217. M appears in 221 and leads 37.

Simulated programmes are saved, to be compared with the real one once a government is invested.